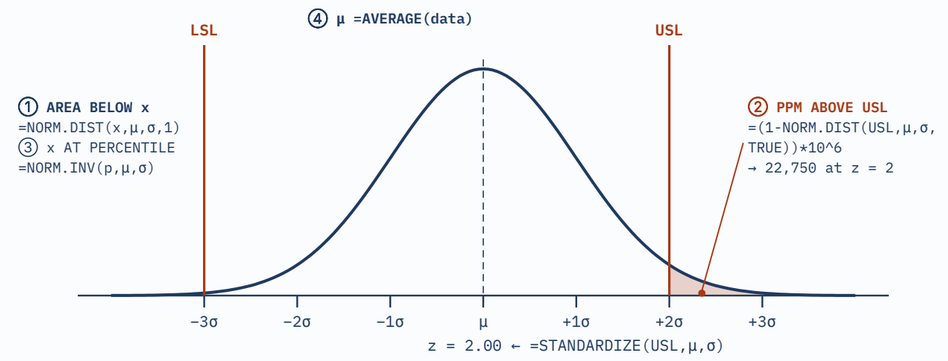
UPSKILL SPRINT · WALL REFERENCE · LEAN SIX SIGMA & QUALITY

EXCEL STATISTICS

GENERAL ARRANGEMENT AND COMPLETE PARTS LIST

All 111 statistical functions, 21 helpers, 8 What-if tools, 19 ToolPak tools, 39 superseded names and 12 gaps, itemised like a drawing. Item numbers match the lesson's map stops.

VIEW A · THE NORMAL CURVE, DIMENSIONED IN EXCEL



PARTS LIST 01 · DESCRIPTIVE STATISTICS — QTY 36

01.01 **AVERAGE**(number1, [number2], ...)
Default centre of a data set

01.02 **AVERAGEA**(value1, [value2], ...)
Only when logical values must count

01.03 **AVERAGEIF**(range, criteria, [average_range])
Mean for one shift, grade or line

01.04 **AVERAGEIFS**(average_range, criteria_range1, criteria1, ...)
Mean by heat and orientation together

01.05 **MEDIAN**(number1, [number2], ...)
Skewed data such as delays and cycle times

01.06 **MODE.SNGL**(number1, [number2], ...)
Discrete readings with repeats

01.07 **MODE.MULT**(number1, [number2], ...)
Data with two or more peaks

01.08 **GEOMEAN**(number1, [number2], ...)
Growth rates, ratios, log-normal data

01.09 **HARMEAN**(number1, [number2], ...)
Averaging rates such as units per hour

01.10 **TRIMMEAN**(array, percent)
A robust mean when outliers are present

01.11 **STDEV.S**(number1, [number2], ...)
Almost every quality calculation on sample data

01.12 **STDEV.P**(number1, [number2], ...)
Only when you hold the whole population

01.13 **STDEVA**(value1, [value2], ...)
Rarely; text becomes 0 and distorts results

01.14 **STDEVPA**(value1, [value2], ...)
Rarely; same caution as STDEVA

01.15 **VAR.S**(number1, [number2], ...)
ANOVA, F-tests and pooled variance

01.16 **VAR.P**(number1, [number2], ...)
Complete populations only

01.17 **VARA**(value1, [value2], ...)
Rarely

01.18 **VARPA**(value1, [value2], ...)
Rarely

01.19 **AVEDEV**(number1, [number2], ...)
A spread summary that resists outliers

01.20 **DEVSQ**(number1, [number2], ...)
Building ANOVA sums of squares by hand

01.21 **SKEW**(number1, [number2], ...)
Quick normality and symmetry screen

01.22 **SKEW.P**(number1, [number2], ...)
Skewness of a complete population

01.23 **KURT**(number1, [number2], ...)
Screening for heavy or light tails

01.24 **MAX**(number1, [number2], ...)
Range and worst-case checks

01.25 **MAXA**(value1, [value2], ...)
Rarely

01.26 **MAXIFS**(max_range, criteria_range1, criteria1, ...)
Worst reading per grade or line

01.27 **MIN**(number1, [number2], ...)
Range and minimum-spec checks

01.28 **MINA**(value1, [value2], ...)
Rarely

01.29 **MINIFS**(min_range, criteria_range1, criteria1, ...)
Lowest toughness per heat

01.30 **LARGE**(array, k)
Top-k values and order statistics

01.31 **SMALL**(array, k)
Bottom-k values and order statistics

01.32 **COUNT**(value1, [value2], ...)
The sample size for every other formula

01.33 **COUNTA**(value1, [value2], ...)
Checking for text entries in numeric columns

01.34 **COUNTBLANK**(range)
Missing-data audit

01.35 **COUNTIF**(range, criteria)
Defects beyond a spec limit

01.36 **COUNTIFS**(criteria_range1, criteria1, ...)
Contingency tables and DPU by category

PARTS LIST 02 · PROBABILITY DISTRIBUTIONS — QTY 41

02.01 **NORM.DIST**(x, mean, standard_dev, cumulative)
Share of output below a limit, ppm out of spec

02.02 **NORM.INV**(probability, mean, standard_dev)
Percentile limits; simulation with RAND()

02.03 **NORM.S.DIST**(z, cumulative)
p-values for z statistics

02.04 **NORM.S.INV**(probability)
Critical z, sample size, sigma level

02.05 **GAUSS**(z)
Quick area from the centre

02.06 **PHI**(x)
Plotting the bell curve

02.07 **T.DIST**(x, deg_freedom, cumulative)
Lower-tailed t tests

02.08 **T.DIST.RT**(x, deg_freedom)
Upper-tailed t tests

02.09 **T.DIST.2T**(x, deg_freedom)
Two-sided p-value for a t statistic

02.10 **T.INV**(probability, deg_freedom)
One-sided critical values; negative for small p

02.11 **T.INV.2T**(probability, deg_freedom)
Confidence intervals and two-sided tests

02.12 **CHISQ.DIST**(x, deg_freedom, cumulative)
Lower tail of a variance test

02.13 **CHISQ.DIST.RT**(x, deg_freedom)
p-value for chi-square tests with any df

02.14 **CHISQ.INV**(probability, deg_freedom)
Upper limit of a variance confidence interval

02.15 **CHISQ.INV.RT**(probability, deg_freedom)
Chi-square table lookups; variance intervals

02.16 **F.DIST**(x, deg_freedom1, deg_freedom2, cumulative)
Lower tail of a variance-ratio test

02.17 **F.DIST.RT**(x, deg_freedom1, deg_freedom2)
ANOVA and regression p-values

02.18 **F.INV**(probability, deg_freedom1, deg_freedom2)
Lower limit of a variance-ratio interval

02.19 **F.INV.RT**(probability, deg_freedom1, deg_freedom2)
F tables for ANOVA

02.20 **BINOM.DIST**(number_s, trials, probability_s, cumulative)
Defectives in a sample, pass or fail data

02.21 **BINOM.DIST.RANGE**(trials, probability_s, number_s, [number_s2])
Acceptance-sampling ranges in one step

02.22 **BINOM.INV**(trials, probability_s, alpha)
Acceptance numbers and exact critical counts

02.23 **NEGBINOM.DIST**(number_f, number_s, probability_s, cumulative)
Inspections until a set number of finds

02.24 **HYPGEOM.DIST**(sample_s, number_sample, population_s, number_pop, cumulative)
Sampling small lots without replacement

02.25 **POISSON.DIST**(x, mean, cumulative)
Defects per unit, c- and u-chart probabilities

02.26 **EXPON.DIST**(x, lambda, cumulative)
Time between failures, constant failure rate

02.27 **WEIBULL.DIST**(x, alpha, beta, cumulative)
Life data and wear-out reliability

02.28 **LOGNORM.DIST**(x, mean, standard_dev, cumulative)
Repair times, particle sizes, skewed data

02.29 **LOGNORM.INV**(probability, mean, standard_dev)
Percentile limits for skewed data

02.30 **GAMMA.DIST**(x, alpha, beta, cumulative)
Waiting times for several events

02.31 **GAMMA.INV**(probability, alpha, beta)
Percentiles of gamma-shaped data

02.32 **GAMMA**(number)
Building other distributions by hand

02.33 **GAMMALN**(x)
Large factorials without overflow

02.34 **GAMMALN.PRECISE**(x)
Same use, newer algorithm

02.35 **BETA.DIST**(x, alpha, beta, cumulative, [A], [B])
PERT task times, proportions

02.36 **BETA.INV**(probability, alpha, beta, [A], [B])
Exact binomial confidence limits

02.37 **PERMUT**(number, number_chosen)
Counting sequences and orders

02.38 **PERMUTATIONA**(number, number_chosen)
Codes and passwords style counting

02.39 **PROB**(x_range, prob_range, [lower_limit], [upper_limit])
Custom discrete distributions

02.40 **COMBIN**(number, number_chosen) HELPER
Counting samples where order does not matter

02.41 **COMBINA**(number, number_chosen) HELPER
Counting with replacement

PARTS LIST 03 · STATISTICAL INFERENCE — QTY 2

03.01 **CONFIDENCE.T**(alpha, standard_dev, size)
Confidence interval for a mean, sigma unknown

03.02 **CONFIDENCE.NORM**(alpha, standard_dev, size)
Confidence interval for a mean, sigma known

PARTS LIST 04 · HYPOTHESIS TESTS — QTY 4

04.01 **T.TEST**(array1, array2, tails, type)
Comparing two means

04.02 **Z.TEST**(array, x, [sigma])
Mean against a target when sigma is known

04.03 **F.TEST**(array1, array2)
Checking variances before choosing T.TEST type

04.04 **CHISQ.TEST**(actual_range, expected_range)
Counts by category; contingency tables

PARTS LIST 05 · CORRELATION — QTY 7

05.01 **CORREL**(array1, array2)
Strength and direction of a straight-line link

05.02 **PEARSON**(array1, array2)
Legacy sheets; identical result

05.03 **RSQ**(known_y's, known_x's)
Share of variation explained by one x

05.04 **COVARIANCE.S**(array1, array2)
Building r by hand on sample data

05.05 **COVARIANCE.P**(array1, array2)
Matches the ToolPak Covariance tool

05.06 **FISHER**(x)
Confidence interval on r; comparing two r values

05.07 **FISHERINV**(y)
Second half of the r interval

PARTS LIST 06 · REGRESSION — QTY 7

06.01 **LINEST**(known_y's, [known_x's], [const], [stats])
Simple or multiple regression with inference

06.02 **SLOPE**(known_y's, known_x's)
Change in y per unit of x

06.03 **INTERCEPT**(known_y's, known_x's)
y when x is zero

06.04 **STEYX**(known_y's, known_x's)
Width of prediction intervals

06.05 **TREND**(known_y's, [known_x's], [new_x's], [const])
Fitted values, residuals, predictions

06.06 **LOGEST**(known_y's, [known_x's], [const], [stats])
Growth or decay curves

06.07 **GROWTH**(known_y's, [known_x's], [new_x's], [const])
Predictions from an exponential model

PARTS LIST 07 · FORECASTING — QTY 6

07.01 **FORECAST.LINEAR**(x, known_y's, known_x's)
Straight-line trend forecasts

07.02 **FORECAST**(x, known_y's, known_x's)
Older workbooks only

07.03 **FORECAST.ETS**(target_date, values, timeline, [seasonality], [data_completion], [aggregation])
Time series with trend and seasonality

07.04 **FORECAST.ETS.CONFINT**(target_date, values, timeline, [confidence_level], [seasonality], [data_completion], [aggregation])
Error bands around ETS forecasts

07.05 **FORECAST.ETS.SEASONALITY**(values, timeline, [data_completion], [aggregation])
Confirming the cycle before trusting a forecast

07.06 **FORECAST.ETS.STAT**(values, timeline, statistic_type, [seasonality], [data_completion], [aggregation])
Judging forecast accuracy

PARTS LIST 08 · EXPLORATORY DATA ANALYSIS — QTY 18

08.01 **FREQUENCY**(data_array, bins_array)
Histograms and goodness-of-fit counts

08.02 **QUARTILE.EXC**(array, quart)
Box plots that match Minitab

08.03 **QUARTILE.INC**(array, quart)
Matching older Excel workbooks

08.04 **PERCENTILE.EXC**(array, k)
P5 or P95 limits that match Minitab

08.05 **PERCENTILE.INC**(array, k)
Percentiles that include 0 and 1

08.06 **PERCENTRANK.EXC**(array, x, [significance])
Where one value sits in the data

08.07 **PERCENTRANK.INC**(array, x, [significance])
Same, inclusive method

08.08 **RANK.AVG**(number, ref, [order])
Normal probability plots and rank tests

08.09 **RANK.EQ**(number, ref, [order])
League tables and Pareto ordering

08.10 **STANDARDIZE**(x, mean, standard_dev)
Flagging points beyond three standard deviations

08.11 **GROUPBY**(row_fields, values, function, [field_headers], ...) HELPER
Stratify by heat, shift, line or supplier

08.12 **PIVOTBY**(row_fields, col_fields, values, function, ...) HELPER
Side-by-side comparisons such as LPA vs TWA

08.13 **PERCENTOF**(data_subset, data_all) HELPER
Pareto-style contributions

08.14 **FILTER**(array, include, [if_empty]) HELPER
Slicing a subgroup without copying

08.15 **UNIQUE**(array, [by_col], [exactly_once]) HELPER
Listing categories before grouping

08.16 **SORT**(array, [sort_index], [sort_order], [by_col]) HELPER
Ordering data for plots and ranks

08.17 **SUBTOTAL**(function_num, ref1, [ref2], ...) HELPER
Summaries that follow a filter

08.18 **AGGREGATE**(function_num, options, ref1, [ref2], ...) HELPER
Robust summaries on messy data

PARTS LIST 09 · WHAT-IF & OPTIMISATION — QTY 18

09.01 **Goal Seek** · Data › What-If Analysis › Goal Seek
Which sigma gives Cpk 1.33?

09.02 **Data Table (one variable)** · Data › What-If Analysis › Data Table (column input cell)
Sensitivity of cost or yield to one input

09.03 **Data Table (two variables)** · Data › What-If Analysis › Data Table (row and column input cells)
Cpk map across mean and sigma

09.04 **Scenario Manager** · Data › What-If Analysis › Scenario Manager
Best, likely and worst cases

09.05 **Solver (Simplex LP)** · Data › Solver › Select a Solving Method: Simplex LP
Blends, product mix, scheduling

09.06 **Solver (GRG Nonlinear)** · Data › Solver › Select a Solving Method: GRG Nonlinear
Response optimization, curve fitting

09.07 **Solver (Evolutionary)** · Data › Solver › Select a Solving Method: Evolutionary
Models with IF, LOOKUP or rounding

09.08 **Forecast Sheet** · Data › Forecast › Forecast Sheet
One-click seasonal forecast

09.09 **RAND()** HELPER
Driving simulations through .INV functions

09.10 **RANDARRAY**([rows], [columns], [min], [max], [integer]) HELPER
Thousands of draws in one formula

09.11 **RANDBETWEEN**(bottom, top) HELPER
Random sampling of rows

09.12 **SEQUENCE**(rows, [columns], [start], [step]) HELPER
Input grids for sensitivity tables

09.13 **LET**(name1, name_value1, calculation_or_name2, ...) HELPER
Readable one-cell simulations

09.14 **LAMBDA**([parameter1, ...], calculation) HELPER
Custom test statistics and models

09.15 **MAP**(array1, lambda_or_array) HELPER
Formula-based data tables that recalculate fast

09.16 **SUMPRODUCT**(array1, [array2], ...) HELPER
Objective and constraint cells for Solver

09.17 **MMULT**(array1, array2) HELPER
Regression by matrices; linear models

09.18 **MINVERSE**(array) HELPER
Normal equations for regression by hand

PARTS LIST 10 · ANALYSIS TOOLPAK — QTY 19

10.01 **Anova: Single Factor** · Data › Data Analysis › Anova: Single Factor
Three or more group means

10.02 **Anova: Two-Factor With Replication** · Data › Data Analysis › Anova: Two-Factor With Replication
Two factors with repeats

10.03 **Anova: Two-Factor Without Replication** · Data › Data Analysis › Anova: Two-Factor Without Replication
Randomized block designs

10.04 **t-Test: Paired Two Sample for Means** · Data › Data Analysis › t-Test: Paired Two Sample for Means
Before and after on the same units

10.05 **t-Test: Two-Sample Assuming Equal Variances** · Data › Data Analysis › t-Test: Two-Sample Assuming Equal Variances
Independent groups, equal spread

10.06 **t-Test: Two-Sample Assuming Unequal Variances** · Data › Data Analysis › t-Test: Two-Sample Assuming Unequal Variances
Independent groups, unequal spread

10.07 **z-Test: Two Sample for Means** · Data › Data Analysis › z-Test: Two Sample for Means
Two means with known variances

10.08 **F-Test Two-Sample for Variances** · Data › Data Analysis › F-Test Two-Sample for Variances
Comparing two variances

10.09 **Descriptive Statistics** · Data › Data Analysis › Descriptive Statistics
One-click summary

10.10 **Regression** · Data › Data Analysis › Regression
Full regression output

10.11 **Correlation** · Data › Data Analysis › Correlation
Several variables at once

10.12 **Covariance** · Data › Data Analysis › Covariance
Several variables at once

10.13 **Histogram** · Data › Data Analysis › Histogram
Quick frequency table

10.14 **Rank and Percentile** · Data › Data Analysis › Rank and Percentile
Relative standing table

10.15 **Random Number Generation** · Data › Data Analysis › Random Number Generation
Static simulated data

10.16 **Sampling** · Data › Data Analysis › Sampling
Drawing an audit sample

10.17 **Moving Average** · Data › Data Analysis › Moving Average
Trend smoothing

10.18 **Exponential Smoothing** · Data › Data Analysis › Exponential Smoothing
Level-only forecasts

10.19 **Fourier Analysis** · Data › Data Analysis › Fourier Analysis
Vibration and periodic signals

PARTS LIST 11 · LEGACY NAMES — QTY 39

11.01 ~~**BETADIST**~~ → **BETA.DIST**
11.02 ~~**BETAINV**~~ → **BETA.INV**
11.03 ~~**BINOMDIST**~~ → **BINOM.DIST**
11.04 ~~**CHIDIST**~~ → **CHISQ.DIST.RT**
11.05 ~~**CHIINV**~~ → **CHISQ.INV.RT**
11.06 ~~**CHITEST**~~ → **CHISQ.TEST**
11.07 ~~**CONFIDENCE**~~ → **CONFIDENCE.NORM**
11.08 ~~**COVAR**~~ → **COVARIANCE.P**
11.09 ~~**CRITBINOM**~~ → **BINOM.INV**
11.10 ~~**EXPONDIST**~~ → **EXPON.DIST**
11.11 ~~**FDIST**~~ → **F.DIST.RT**
11.12 ~~**FINV**~~ → **F.INV.RT**
11.13 ~~**FORECAST**~~ → **FORECAST.LINEAR**
11.14 ~~**FTEST**~~ → **F.TEST**
11.15 ~~**GAMMADIST**~~ → **GAMMA.DIST**
11.16 ~~**GAMMAINV**~~ → **GAMMA.INV**
11.17 ~~**HYPGEOMDIST**~~ → **HYPGEOM.DIST**
11.18 ~~**LOGINV**~~ → **LOGNORM.INV**
11.19 ~~**LOGNORMDIST**~~ → **LOGNORM.DIST**
11.20 ~~**MODE**~~ → **MODE.SNGL**
11.21 ~~**NEGBINOMDIST**~~ → **NEGBINOM.DIST**
11.22 ~~**NORMDIST**~~ → **NORM.DIST**
11.23 ~~**NORMINV**~~ → **NORM.INV**
11.24 ~~**NORMSDIST**~~ → **NORM.S.DIST**
11.25 ~~**NORMSINV**~~ → **NORM.S.INV**
11.26 ~~**PERCENTILE**~~ → **PERCENTILE.INC**
11.27 ~~**PERCENTRANK**~~ → **PERCENTRANK.INC**
11.28 ~~**POISSON**~~ → **POISSON.DIST**
11.29 ~~**QUARTILE**~~ → **QUARTILE.INC**
11.30 ~~**RANK**~~ → **RANK.EQ**
11.31 ~~**STDEV**~~ → **STDEV.S**
11.32 ~~**STDEVP**~~ → **STDEV.P**
11.33 ~~**TDIST**~~ → **T.DIST.RT or T.DIST.2T**
11.34 ~~**TINV**~~ → **T.INV.2T**
11.35 ~~**TTEST**~~ → **T.TEST**
11.36 ~~**VAR**~~ → **VAR.S**
11.37 ~~**VARP**~~ → **VAR.P**
11.38 ~~**WEIBULL**~~ → **WEIBULL.DIST**
11.39 ~~**ZTEST**~~ → **Z.TEST**

PARTS LIST 12 · BEYOND EXCEL — QTY 12

12.01 **Normality test (Anderson-Darling, Shapiro-Wilk)**
Excel workaround: Normal probability plot with NORM.S.INV and RANK.AVG; check SKEW and KURT

12.02 **One- and two-proportion tests**
Excel workaround: z statistic with NORM.S.DIST; exact test with BINOM.DIST

12.03 **Nonparametric tests (sign, Wilcoxon, Mann-Whitney, Kruskal-Wallis)**
Excel workaround: Sign test with BINOM.DIST; rank sums with RANK.AVG

12.04 **Equal variances, three or more groups (Levene, Bartlett)**
Excel workaround: One-way ANOVA on absolute deviations from each group median

12.05 **Post-hoc comparisons (Tukey)**
Excel workaround: Fisher LSD with T.INV.2T; no studentized range in Excel

12.06 **Unbalanced or multi-factor ANOVA and DOE analysis**
Excel workaround: LINEST with coded columns for two-level factorials

12.07 **Power and sample size**
Excel workaround: Sample-size formula with NORM.S.INV

12.08 **Process capability**
Excel workaround: Cpk from AVERAGE and STDEV.S, or R-bar over d2

12.09 **Control charts**
Excel workaround: Limits from AVERAGE and constants, drawn as a line chart

12.10 **Gage R&R**
Excel workaround: ANOVA method built from DEVSQ

12.11 **Weibull fitting**
Excel workaround: LINEST on the linearized Weibull plot

12.12 **Logistic or Poisson regression**
Excel workaround: Solver maximizing the log-likelihood

12.13 **PY**(python_code, return_type) HELPER
scipy.stats and statsmodels for tests Excel lacks

GENERAL NOTES

1. Functions per Microsoft 365 Excel. Item 11 names still calculate; do not specify on new work.
2. Use .S functions for samples. STDEV.S is overall σ: Ppk, not Cpk.
3. p-values two-tailed unless noted. Z.TEST returns the upper one-tail p.
4. QUARTILE.EXC matches Minitab quartiles.
5. Parts list 10 requires the Analysis ToolPak add-in.
6. No native normality test: see item 12.01.

TITLE: EXCEL STATISTICS · GA + PARTS LIST			
DRAWING NO.: USS-XL-STAT-001	REV: B	SCALE: NTS	SHEET: 1 OF 1
DRAWN: UPSKILL SPRINT	WEB: UPSKILLSPRINT.COM		